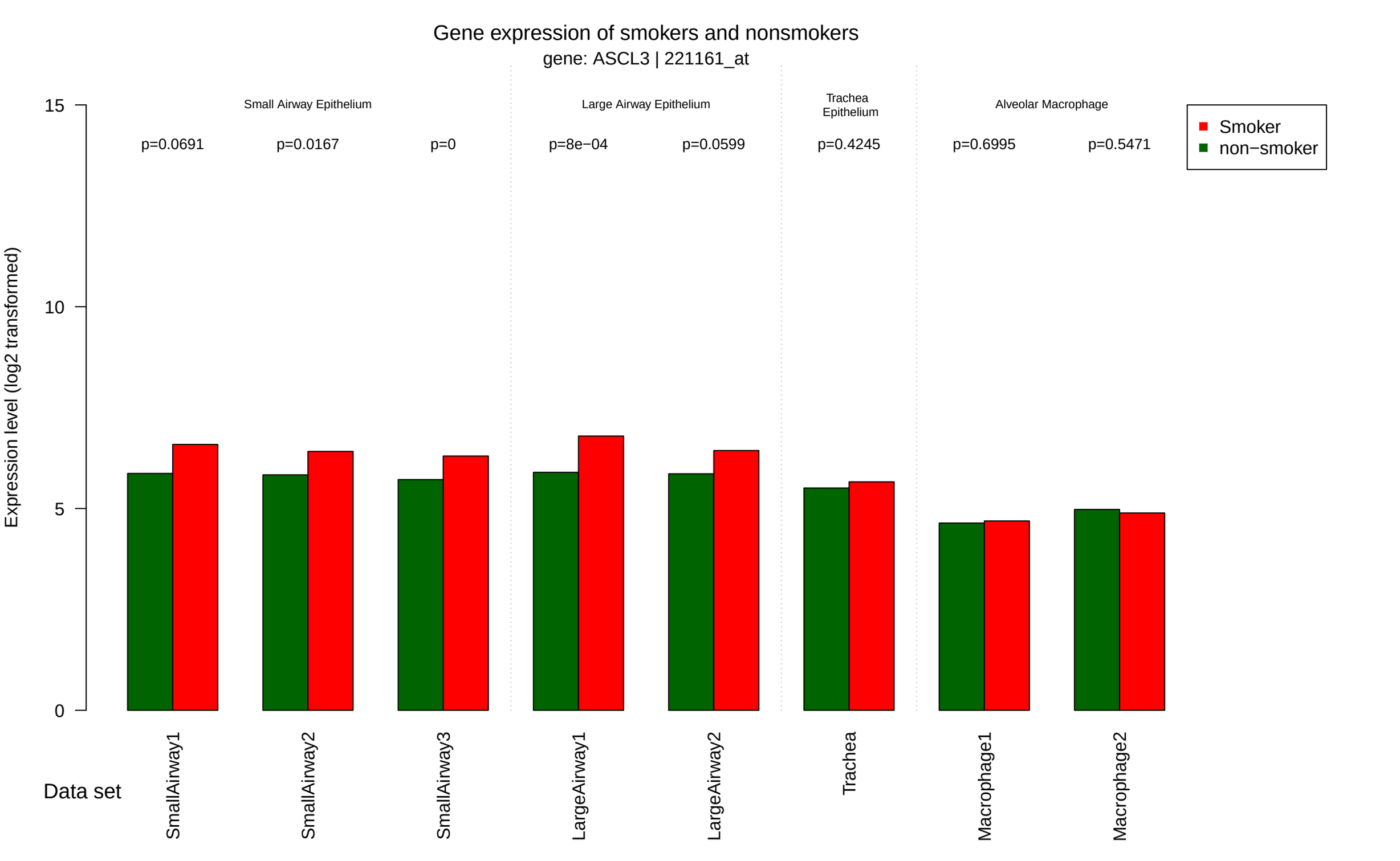
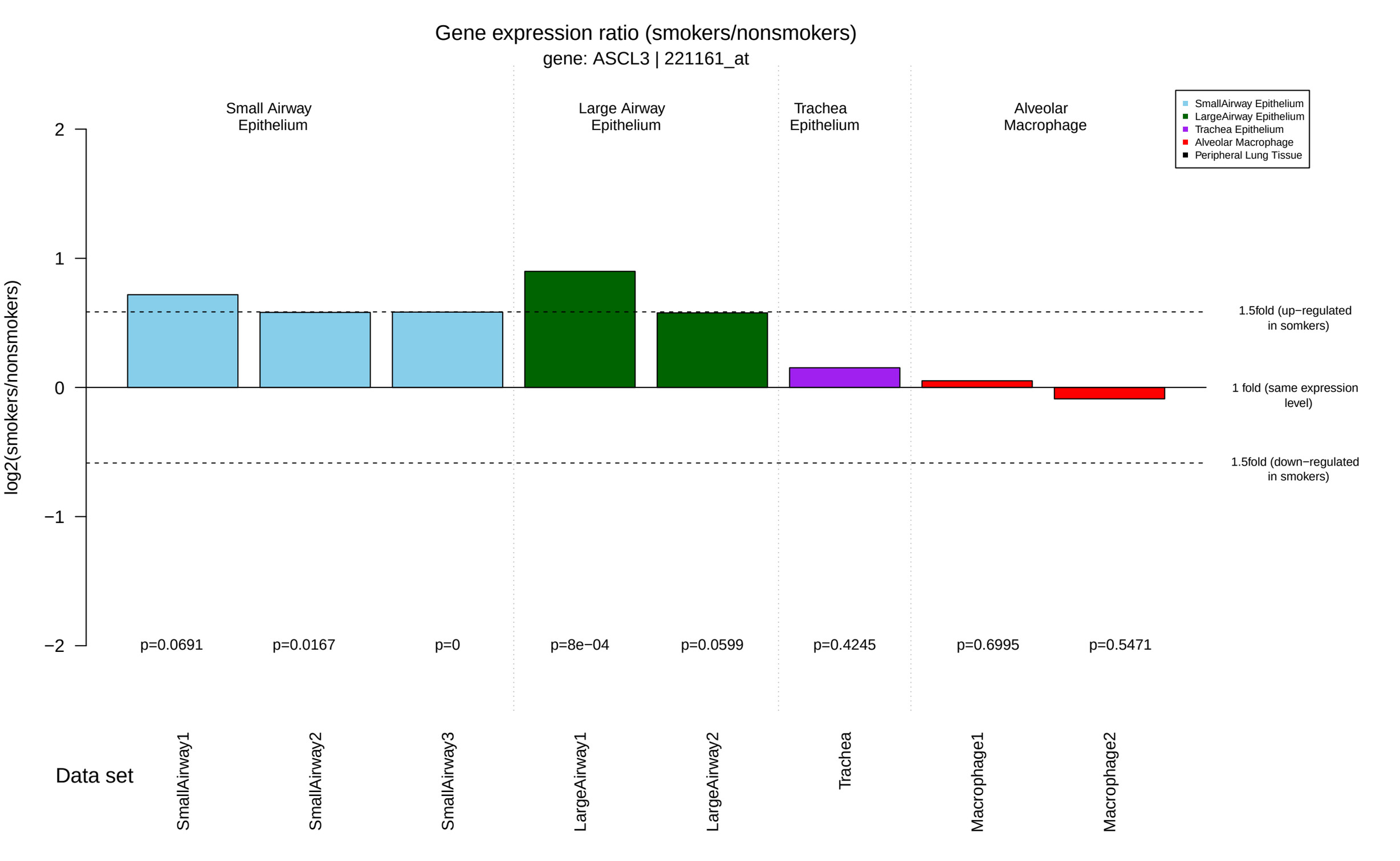
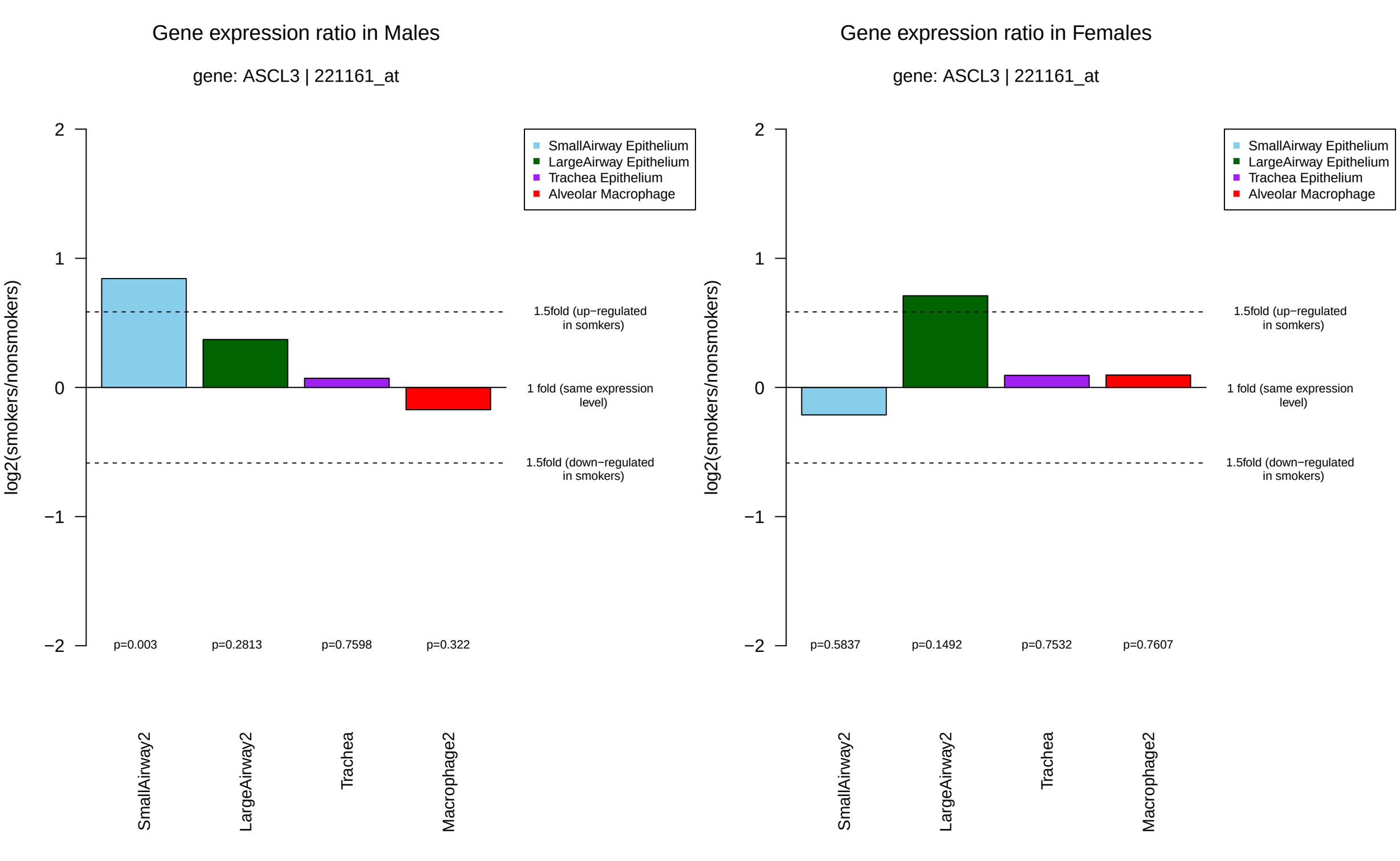
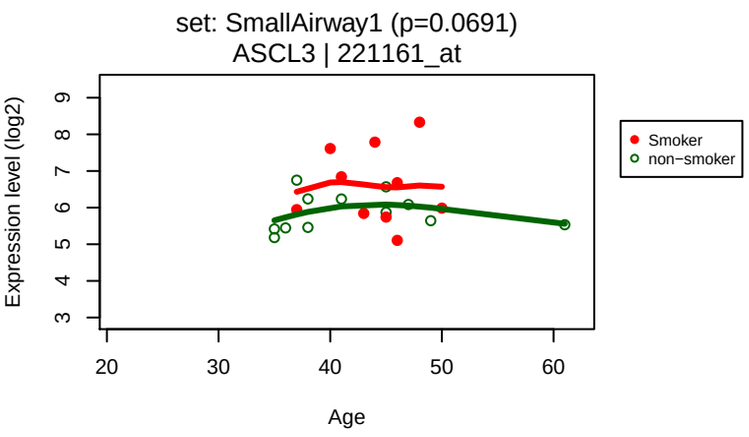
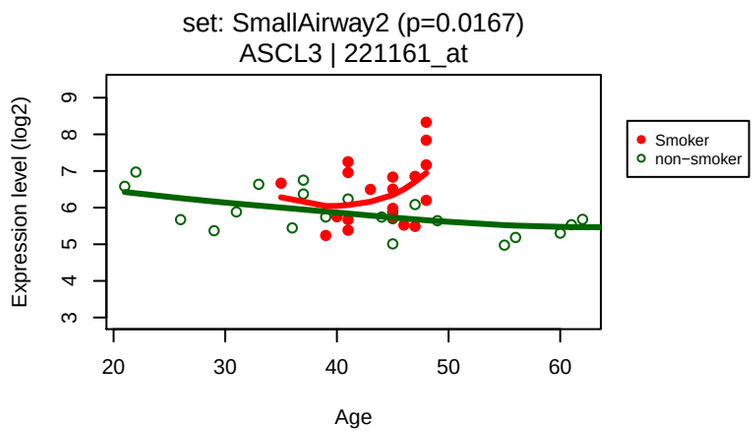
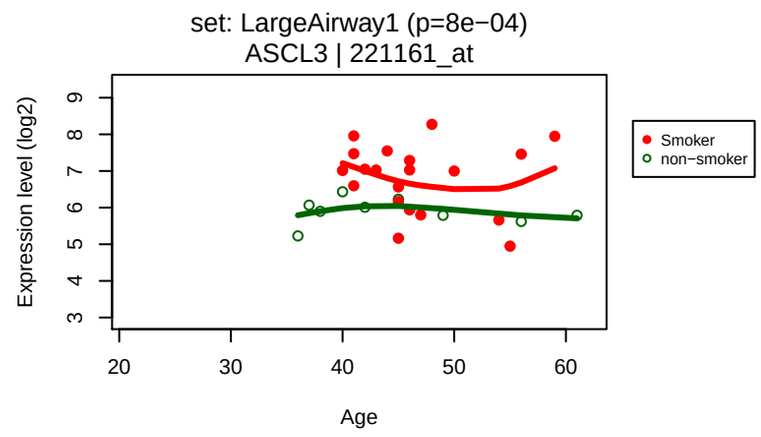

Gene expression of smokers and nonsmokers
gene: ASCL3 | 221161_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0691
p=0.0167
p=0
p=8e−04
p=0.0599
p=0.4245
p=0.6995
p=0.5471
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

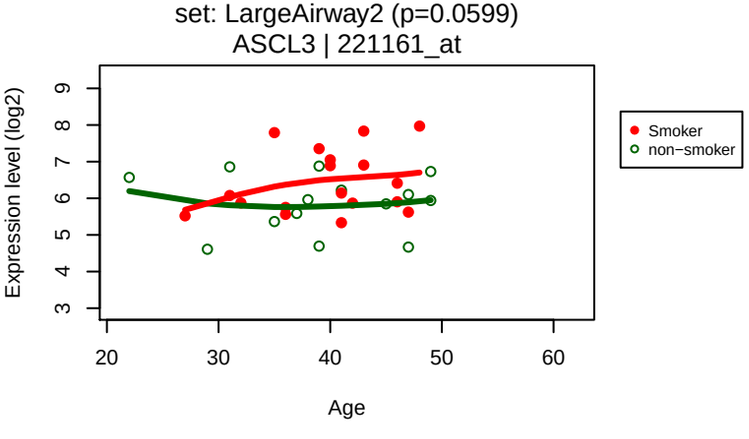

Gene expression ratio (smokers/nonsmokers)
gene: ASCL3 | 221161_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0691
p=0.0167
p=0
p=8e−04
p=0.0599
p=0.4245
p=0.6995
p=0.5471
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

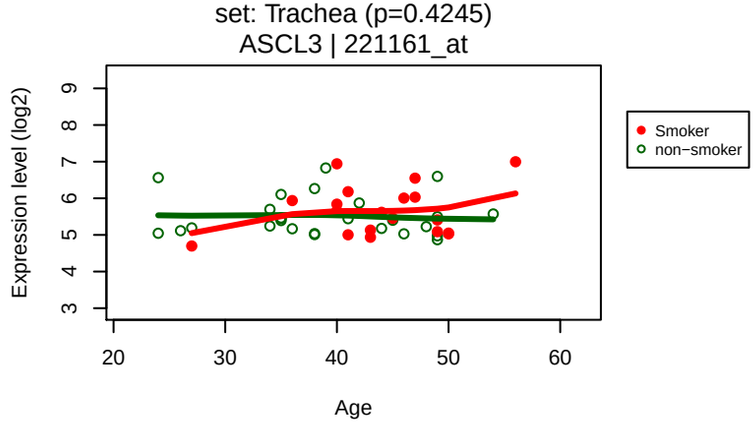

Gene expression ratio in Males
gene: ASCL3 | 221161_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.003
p=0.2813
p=0.7598
p=0.322
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASCL3 | 221161_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5837
p=0.1492
p=0.7532
p=0.7607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

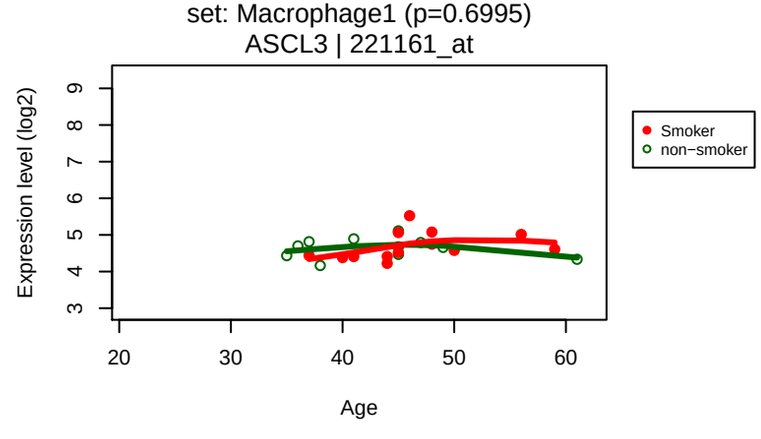

set: SmallAirway1 (p=0.0691)
ASCL3 | 221161_at
Expression level (log2)
Age
Smoker
non−smoker

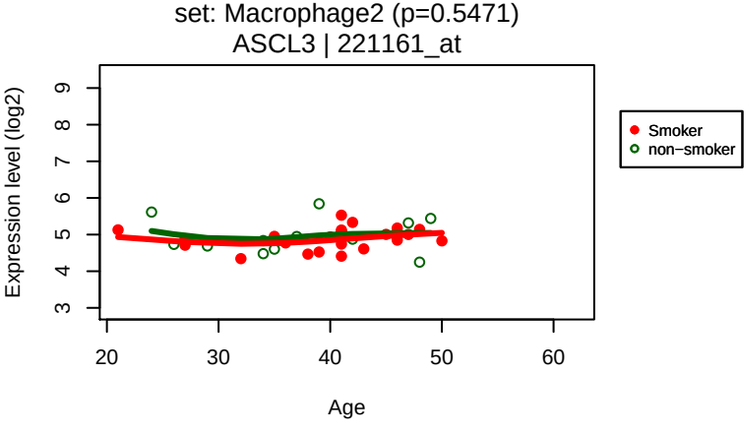

set: SmallAirway2 (p=0.0167)
ASCL3 | 221161_at
Expression level (log2)
Age
Smoker
non−smoker

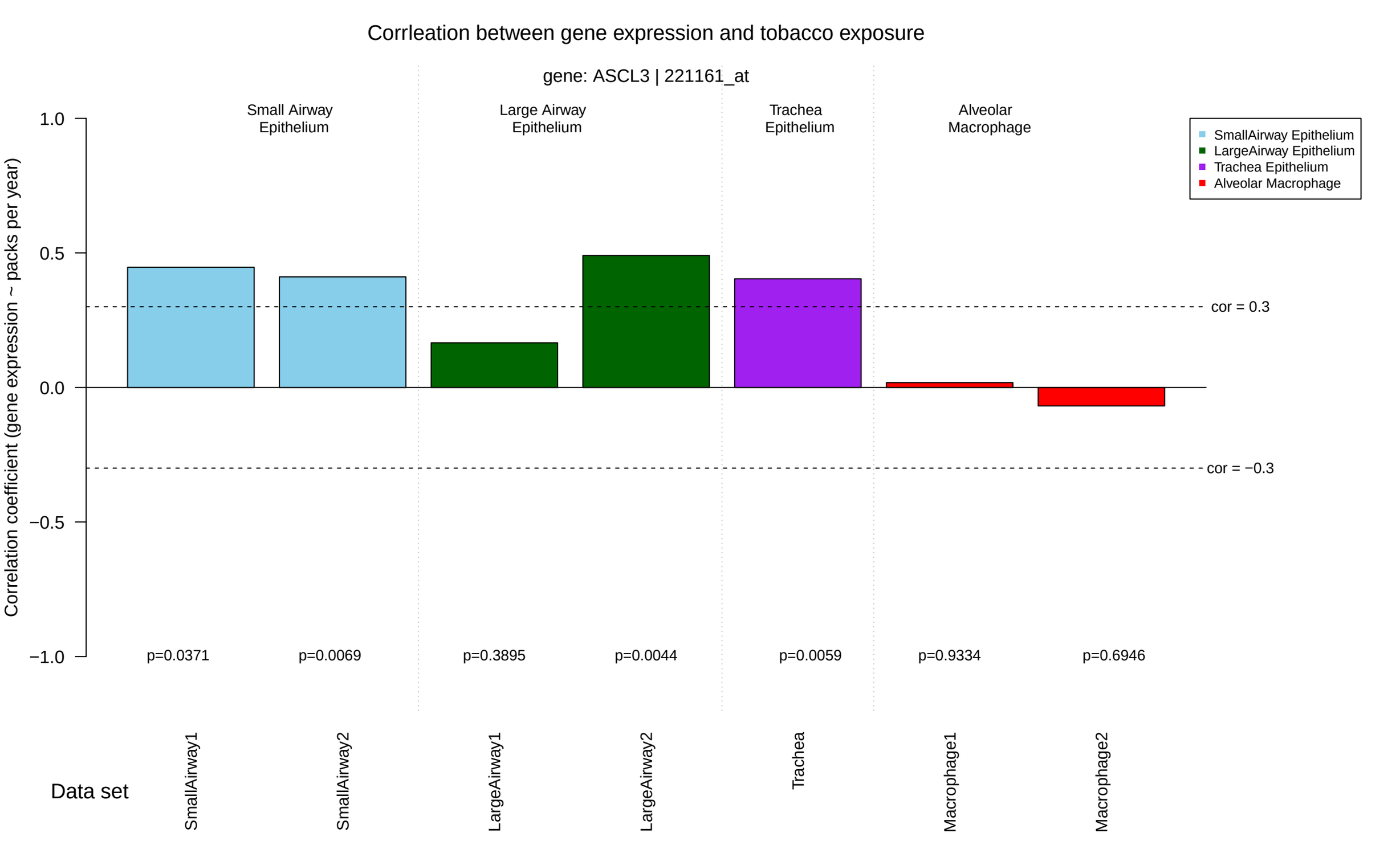

set: LargeAirway1 (p=8e−04)
ASCL3 | 221161_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0599)
ASCL3 | 221161_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4245)
ASCL3 | 221161_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6995)
ASCL3 | 221161_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5471)
ASCL3 | 221161_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASCL3 | 221161_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0371
p=0.0069
p=0.3895
p=0.0044
p=0.0059
p=0.9334
p=0.6946
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2